Notes on parameterization of manipulators via twists, MLS Chapter 3, Section 2.3

On deriving eq. (3.9) p. 94:

One can define the following transformations for the manipulator in the initial configuration:

$$
\begin{aligned}
g_{l_0l_2} &= g_{l_0l_1}g_{l_1l_2} \\
g_{l_0l_3} &= g_{l_0l_1}g_{l_1l_2}g_{l_2l_3} \\
g_{l_0l_4} &= g_{l_0l_1}g_{l_1l_2}g_{l_2l_3}g_{l_3l_4} \\
&\vdots \\
g_{l_0l_{n-1}} &= g_{l_0l_1}g_{l_1l_2}g_{l_2l_3}\dots g_{l_{n-2}l_{n-1}} \\
g_{l_0n} = g_{l_0t} &= g_{l_0l_1}g_{l_1l_2}g_{l_2l_3}\dots g_{l_{n-1}t} = g_{l_0l_1}g_{l_1l_2}g_{l_2l_3}\dots g_{l_{n-1}l_n}
\end{aligned}
$$

where each transformation g on the right hand side of the above equation set is to be interpreted as the one of the g-transformations in the right hand side of eq. (3.8). Then, also,

$$
\begin{aligned}
g^{-1}_{l_0l_2} &= g^{-1}_{l_1l_2}g^{-1}_{l_0l_1} \\
g^{-1}_{l_0l_3} &= g^{-1}_{l_2l_3}g^{-1}_{l_1l_2}g^{-1}_{l_0l_1} \\
g^{-1}_{l_0l_4} &= g^{-1}_{l_3l_4}g^{-1}_{l_2l_3}g^{-1}_{l_1l_2}g^{-1}_{l_0l_1} \\
&\vdots \\
g^{-1}_{l_0l_{n-1}} &= g^{-1}_{l_{n-2}l_{n-1}}\dots g^{-1}_{l_2l_3}g^{-1}_{l_1l_2}g^{-1}_{l_0l_1} \\
g^{-1}_{l_0t} &= g^{-1}_{l_{n-1}t}\dots g^{-1}_{l_2l_3}g^{-1}_{l_1l_2}g^{-1}_{l_0l_1}
\end{aligned}
$$

Eq. (3.8) is re-written here, in red:

$$g_{st}(\theta) = e^{\hat{\xi}_{01}\theta_1}g_{l_0l_1}(0)e^{\hat{\xi}_{12}\theta_2}g_{l_1l_2}(0)\dots e^{\hat{\xi}_{n-2,n-1}\theta_{n-1}}g_{l_{n-2}l_{n-1}}(0)e^{\hat{\xi}_{n-1,n}\theta_n}g_{l_{n-1}l_n}(0)\ .$$

This equation can be re-written, colour-coded, as follows, where the red factors are what were present in the original eq. (3.8), while the black factors were inserted without really changing the equation, because they appear in matrix-inverse-times-matrix pairs:

$$
\begin{aligned}
g_{st}(\theta) \quad = \quad & e^{\hat{\xi}_{01}\theta_1}g_{l_0l_1}(0)e^{\hat{\xi}_{12}\theta_2}g^{-1}_{l_0l_1}(0)g_{l_0l_1}(0)g_{l_1l_2}(0)e^{\hat{\xi}_{23}\theta_3}g^{-1}_{l_1l_2}(0)g^{-1}_{l_0l_1}(0)g_{l_0l_1}(0)g_{l_1l_2}(0)g_{l_2l_3}(0) \\
& \dots e^{\hat{\xi}_{n-2,n-1}\theta_{n-1}}g^{-1}_{l_{n-3}l_{n-2}}(0)\dots g^{-1}_{l_2l_3}(0)g^{-1}_{l_1l_2}(0)g^{-1}_{l_0l_1}(0)g_{l_0l_1}(0)g_{l_1l_2}(0)g_{l_2l_3}(0) \\
& \dots g_{l_{n-3}l_{n-2}}(0)g_{l_{n-2}l_{n-1}}(0)e^{\hat{\xi}_{n-1,n}\theta_n}g_{l_{n-2}l_{n-1}}(0)^{-1}\dots g^{-1}_{l_2l_3}(0)g^{-1}_{l_1l_2}(0)g^{-1}_{l_0l_1}(0)g_{l_0l_1}(0) \\
& g_{l_1l_2}(0)g_{l_2l_3}(0)\dots g_{l_{n-2}l_{n-1}}(0)g_{l_{n-1}l_n}(0)
\end{aligned}
$$

The equation set stated above is now used to simplify this:

$$
\begin{aligned}
g_{st}(\theta) \quad = \quad & e^{\hat{\xi}_{01}\theta_1}\left(g_{l_0l_1}(0)e^{\hat{\xi}_{12}\theta_2}g^{-1}_{l_0l_1}(0)\right)\left(g_{l_0l_2}(0)e^{\hat{\xi}_{23}\theta_3}g^{-1}_{l_0l_2}(0)\right)\left(g_{l_0l_3}(0)e^{\hat{\xi}_{34}\theta_4}g^{-1}_{l_0l_3}(0)\right) \\
& \dots\left(g_{l_0l_{n-1}}(0)e^{\hat{\xi}_{n-1,n}\theta_n}g^{-1}_{l_0l_{n-1}}(0)\right)g_{l_0t}(0)\ ,
\end{aligned}
$$

which is the same as eq. (3.9).